



IEETA



Exploring the Effect of Personalized Virtual Reality Serious Game for Stroke Rehabilitation

JS 2025/2026 – Final Presentation



David Palricas – 108780

INTRODUCTION

Why this Project?

- How does **stroke** impacts our society?
- Why **VR** for rehabilitation?



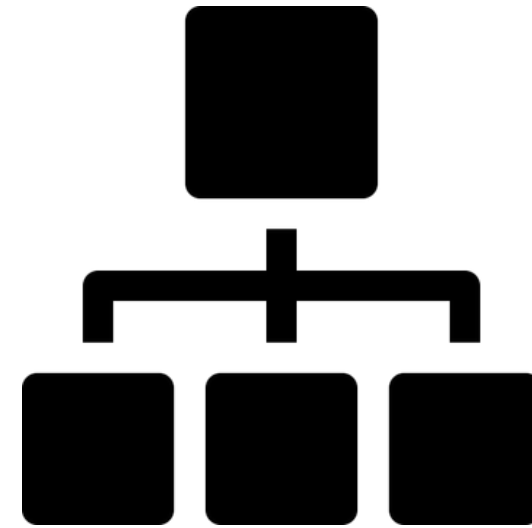
World Stroke
Organization



OBJECTIVES



Create at least 2 VR
mini-games



Create the personalization/
DDA system



Test with users

GAME ENVIRONMENT

Country Fair

- Cultural resonance
- Emotional value
- Promoting an engaging environment



GAME CHARACTERS- FAIR'S OWNER ZECA BIGODES



- Gives an Introduction to the player about the game.
- Congrats The Player after finishing a mini game.

GAME CHARACTERS- CARNY WISE FAIR'S EMPLOYEE

Carny Wise- Fair Employee

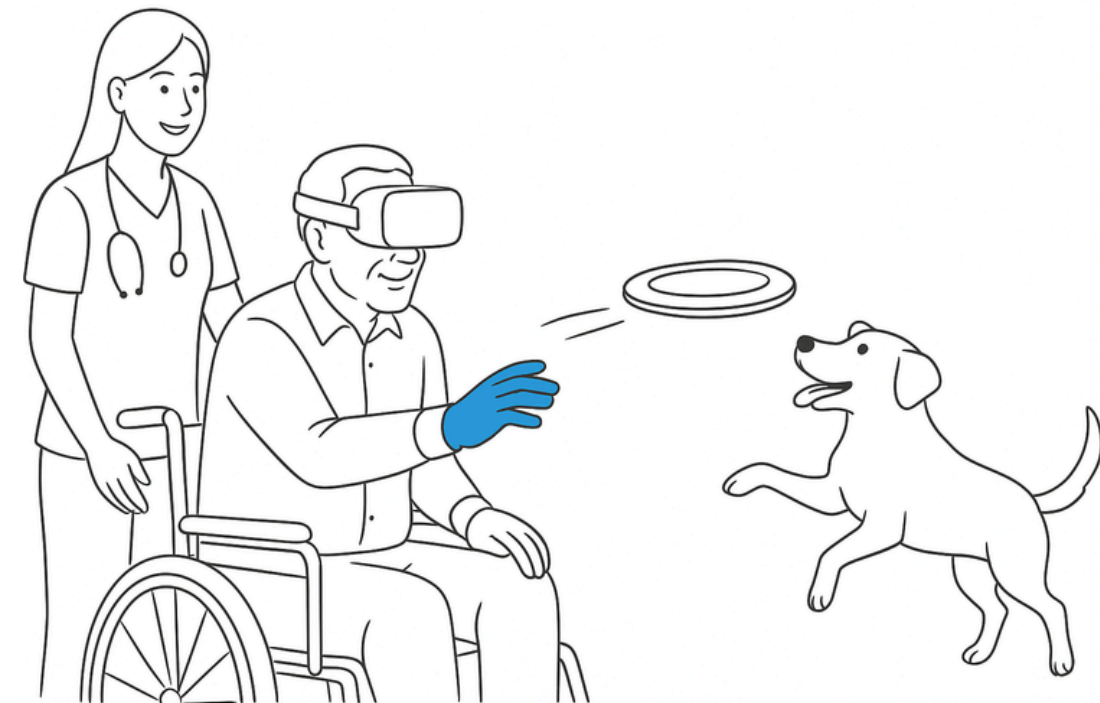


- Explains each mini game rules and goal
- Adjusts the mini games difficulty and notifies when it has changed

MINI GAMES(1/2)

Frisbee Game

- The player must throw a Frisbee to the orange area around the dog or the other orange areas.
- The orange areas can move, disappear or both
- Dog's behavior to catch the frisbee and deliver it to the player
- Score and streak displayed in the UI
- Sound feedback



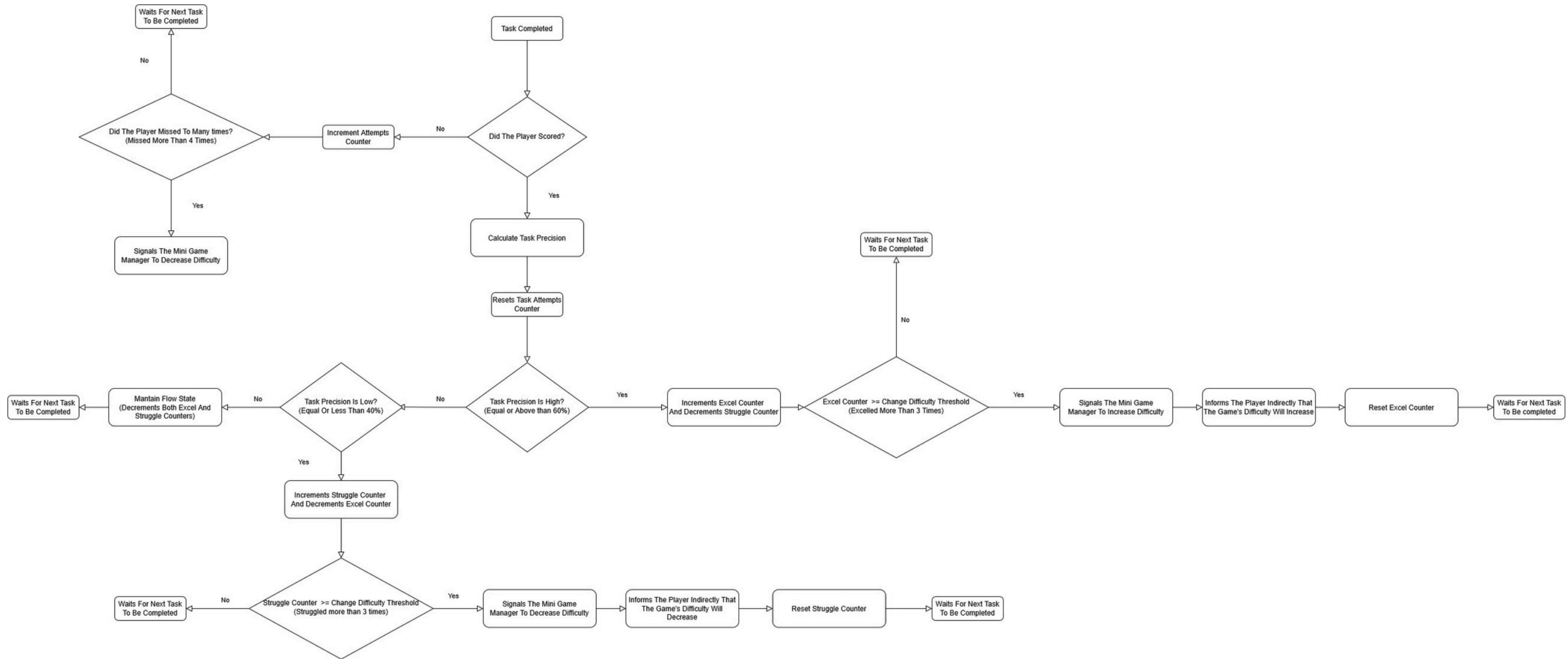
MINI GAMES(2/2)

Archery Game

- The player must shoot balloons with a specific color to score
- Balloons can move, their transparency can change or both.
- Score and streak displayed in the UI
- Sound feedback



DDA SYSTEM- FLOWCHART



DDA SYSTEM- ADAPTIVE PARAMETERS

Parameter Type	Frisbee Game	Archery Game
Target Quantity	<i>Infinite Scaling:</i> Increases by roughly 1 target every 5 levels.	<i>Infinite Scaling:</i> Increases by roughly 1 balloon every 2 levels.
Spatial Challenge	Distance: Linearly increases the dog's distance (capped at 14m).	Density: Increases the number of balloons that the balloon spawner can spawn.
Cognitive Load	Movement: Percentage of targets moving laterally. Visibility: Percentage of targets blinking/disappearing.	Movement: Percentage of balloons moving. Transparency: Percentage of balloons becoming translucent (ghosts). Color Rule: The specific balloon color required to score changes dynamically.

DEMO



TOOLS USED

- Software: Unity, Blender and FMOD
- Hardware: Oculus Quest 3
- Libraries: Meta Interaction SDK, FMOD Unity, DotTween,Newtonsoft



Acknowledgements

- Rita for her work in practical component in Virtual Reality Course
- Google: Sheep Model, Ferris Wheel Model, Tent Model;
- Quaternius: Fence Model;
- J-Toastie: Persons Models and Animations;
- sirkitree: Carnival Booth Model, Merry-Go-Round Model;
- Adam Tomkins: Grass Floor Model;
- Ian MacGillivray: Big Tent Model;
- Don Carson: Popcorn Cart Model;
- Sammy Bench Model.



USER EVALUATION - TASKS

	User 1	User 2
Task 1 Select a mini game to play	12s	3s
Task2 Return to the fair	5s	8s
Task 3 Shoot 3 balloons	11s	11s
Task 4 Score 1 point in Frisbee mini game	53s	37s

USER EVALUATION - FEEDBACK

Overall Experience:

- The mini-games were fun to play
- The users felt they were in a country fair
- Liked the sound feedback
- Felt that the mini-games needed a tutorial

USER EVALUATION - FEEDBACK

Frisbee mini-game:

- The frisbee curved too much in the direction of the throwing hand.
- Instead of going to the user's exact position, the dog delivered the frisbee directly to the nearest outstretched hand

Archery mini-game:

- Balloon Scoring: Considers texture (e.g., metallic) and legend plates, not just color.
- The arrow sometimes launches while the string is still being pulled.

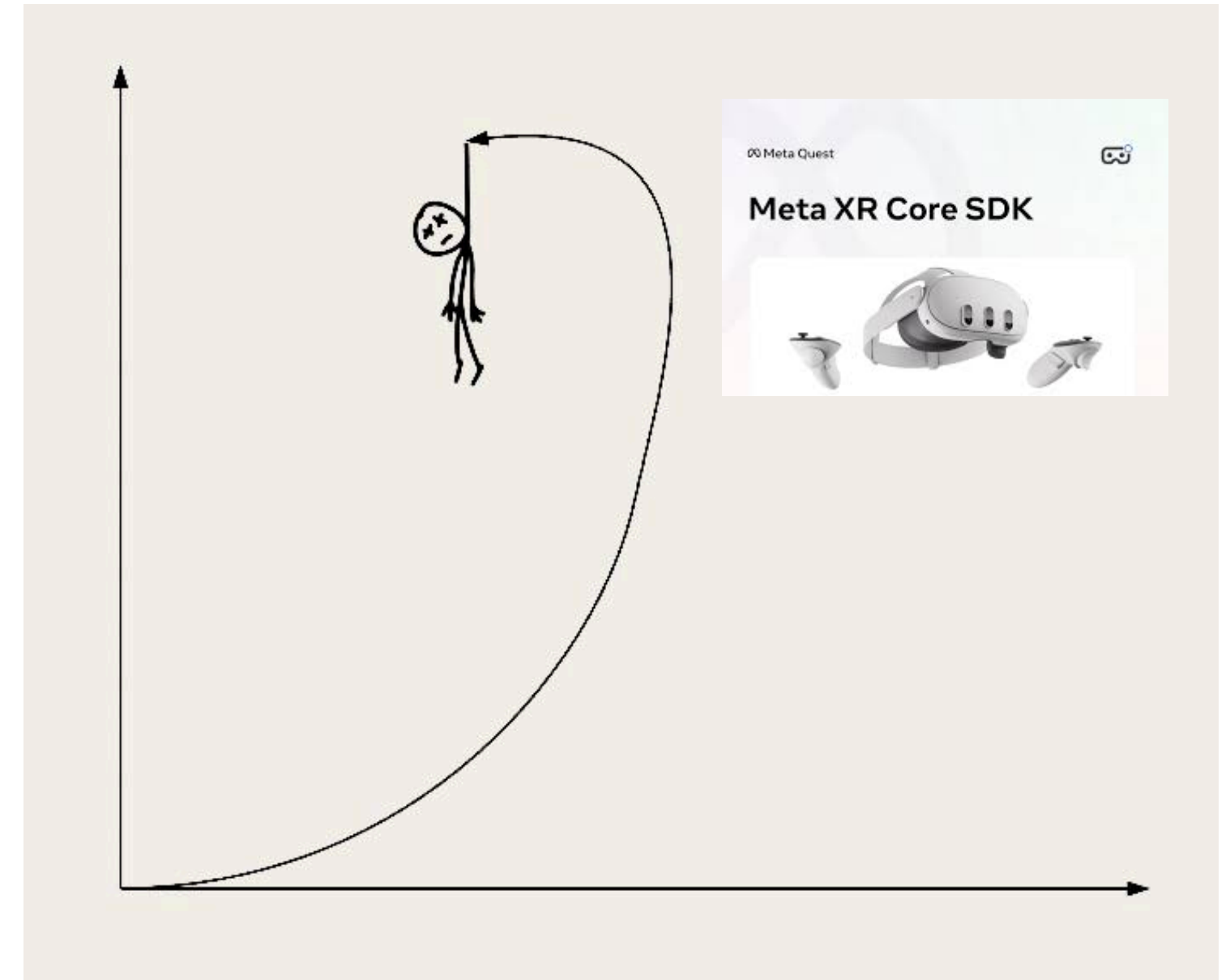
USER EVALUATION - OBSERVATIONS

- Users felt amused while playing
- 1 user struggled to understand how to score a point in the Frisbee mini game
- There were no signs of any negative after effect



MAIN DIFFICULTIES

- Learn VR development in unity
- Hand tracking accuracy and stability
- Implementing and fine-tuning physics
- Debugging and testing in VR
- Android Software problems (Reading JSON files)
- Find information in Meta SDK Documentation



FUTURE WORK

- Improve physics
- Improve the mini game difficulty progression
- Make user tests with target audience

Thank you!